(iii) The value of reciprocal ratios namely cosecant, secant and cotangent can be obtained using the above table.
(iv) The result $\cos 90^\circ = 0$ does not allow us to define $\tan 90^\circ$ and $\sec 90^\circ$.
(v) Similarly $\sin 0^\circ = 0$ does not permit us to define cosec 0° and $\cot 0^\circ$.

3.2.8 Basic Trigonometric Identities

A trigonometric identity represents a relationship that is always true for all admissible values in the domain. For example $\sec\theta = \dfrac{1}{\cos\theta}$ is true for all admissible values of θ. Hence, this is an identity. However, $\sin\theta = \frac{1}{2}$ is not an identity, since the relation fails when $\theta = 60^\circ$. Identities enable us to simplify complicated expressions. They are the basic tools of trigonometry which are being used in solving trigonometric equations. The most important part of working with identities, is to manipulate them with the help of a variety of techniques from algebra.

Let us recall the fundamental identities (Pythagorean identities) of trigonometry, namely,

$$\cos^2\theta + \sin^2\theta = 1$$
$$\sec^2\theta - \tan^2\theta = 1$$
$$\text{cosec}^2\theta - \cot^2\theta = 1$$

(i) $\sin^2\theta$ is the commonly used notation for $(\sin\theta)^2$, likewise for other trigonometric ratios.
(ii) $\sec^2\theta - \tan^2\theta = 1$ is meaningless when $\theta = 90^\circ$. But still it is an identity and true for all values of θ for which $\sec\theta$ and $\tan\theta$ are defined. Thus, an identity is an equation that is true for all values of its domain values.
(iii) When we write $\dfrac{\sin\theta}{1+\cos\theta}$, we understand that the expression is valid for all values of θ for which $1 + \cos\theta \neq 0$.

Example 3.1 Prove that $\dfrac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} = \dfrac{1+\sin\theta}{\cos\theta}$.

Solution:

$$\begin{aligned}\frac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} &= \frac{\tan\theta + \sec\theta - \left(\sec^2\theta - \tan^2\theta\right)}{\tan\theta - \sec\theta + 1} \\ &= \frac{(\tan\theta + \sec\theta)\left[1 - (\sec\theta - \tan\theta)\right]}{\tan\theta - \sec\theta + 1} \\ &= \tan\theta + \sec\theta = \frac{1+\sin\theta}{\cos\theta}.\end{aligned}$$

Example 3.2 Prove that $(\sec A - \text{cosec}A)(1 + \tan A + \cot A) = \tan A \sec A - \cot A\, \text{cosec}A$.

Solution:

$$\text{L.H.S.} = \left(\frac{1}{\cos A} - \frac{1}{\sin A}\right)\left[1 + \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}\right]$$
$$= \frac{\sin^3 A - \cos^3 A}{\sin^2 A \cos^2 A} \quad \text{...(i)}$$
$$\text{R.H.S.} = \frac{\sin A}{\cos^2 A} - \frac{\cos A}{\sin^2 A} = \frac{\sin^3 A - \cos^3 A}{\sin^2 A \cos^2 A} \quad \text{...(ii)}$$

From (i) and (ii), we get the required result.